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Resistance pressure weld for nuclear reactor fuel rod tube end plug

Abstract

A fuel rod for a nuclear reactor, including a cladding tube having a first end with an annular end face, a second end with an annular end face, and a cylindrical body portion extending therebetween, and a first tube end plug including a front portion, an annular lip with an annular end face, and a substantially straight cylindrical body portion extending therebetween, wherein the surface area of the annular end face of the first end of the cladding tube and the annular end face of the annular lip of the first tube end plug are substantially equal, and the annular end face of the first end of the cladding tube and the annular end face of the annular lip of the first tube end plug are connected by a resistance pressure weld.

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Background/Summary

CLAIM OF PRIORITY (1) This application is a division of U.S. patent application Ser. No. 16/562,986 filed on Sep. 6, 2019, now U.S. Pat. No. 11,049,623, which is a division of U.S. application Ser. No. 15/290,628 filed on Oct. 11, 2016, now U.S. Pat. No. 10,410,754, the disclosures of which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to fuel rods for use with fuel assemblies of nuclear reactors. More particularly, the present invention relates to a tube end plug for a fuel rod and a resistance pressure welding method for connecting same to a cladding tube of the fuel rod.

BACKGROUND OF THE INVENTION

(2) In a typical nuclear reactor, the reactor core includes a number of fuel assemblies, each of which is made up of an array of fuel rods **2**, as shown in FIG. **10**. Each fuel rod **2** often includes a tubular cladding **4** containing fuel pellets (not shown) composed of fissile material, and the cladding tube is sealed by upper and lower tube end caps or plugs **6**. The nuclear reactor core is

made up of an array of such fuel rods 2, and is disposed in a pressure vessel containing primary coolant (typically water, although heavy water or another coolant is also contemplated). The primary coolant flows through the nuclear reactor core and is heated by the radioactive core. In a typical boiling water reactor (BWR) configuration, the heated coolant boils to form primary coolant steam that is piped out of the pressure vessel and used to drive one or more turbines. In a typical pressurized water reactor (PWR) configuration, the primary coolant remains in a subcooled state and is piped through steam generators located outside of the vessel to heat secondary coolant that drives one or more turbines. In a variant integral PWR configuration, the steam generators are located inside the pressure vessel and the secondary coolant is pumped into the steam generators.

(3) As noted, each fuel rod 2 typically includes a column of nuclear fuel pellets loaded into a cladding tube 4, and tube end plugs 6 secured to the opposite ends of the cladding tube 4. The end plugs 6 should provide a reliable seal to prevent leakage of primary coolant into the fuel rods 2. In known approaches, the top and bottom end plugs 6 are girth or butt welded to the opposite ends of the tube 4, for example by fusion welding or solid state welding.

(4) Resistance welding of tube end plugs 6 to cladding tubes 4 is also known wherein the cladding tube 4 is butted against the end plug 6 and a controlled high current, typically alternating current (AC), is passed between the cladding tube 4 and the end plug 6 which is compressively loaded. Resistance at the interface between the end plug 6 and the cladding tube 4 generates localized heating resulting in a diffusion bond. While resistance welding has many desirable attributes, the process has some shortcomings. For example, for those nuclear reactors that operate at “high temperatures” (500° C. to 950° C.), the use of special materials, such as martensitic stainless steel, oxide dispersion strengthened alloys (ODS), etc., may be required for construction of the fuel rod's cladding tube and tube end plugs. Although those materials provide enhanced performance characteristics with regard to operation of the fuel rods, they are known to be difficult to weld resulting in undesirable weld properties that may lead to weld failures at an unacceptable rate. Often, because the mass of the tube end plug in the vicinity of the weld region is greater than that of the cladding tube, a large temperature differential between the tube and end plug may lead to degraded bond quality. In order to match the mass of the tube end plug 6 to that of the cladding tube 4 in the vicinity of the weld region, it is known to provide a cylindrical recess 7 in the end of the tube end plug 6 so that the wall thickness are more evenly matched. However, a sharp angular transition 8 from the side wall to the bottom wall of such a recess 7 can lead to stress concentrations and, therefore, reduced strength of the weld region.

(5) Moreover, AC welding has some additional drawbacks. For instance, welding with alternating current causes heating and cooling during each cycle. This can be an important distinction for materials requiring longer welding times (>~16 ms). When welding with AC, weld time is increased by increasing the number of AC cycles. Within these cycles the material cools, subsequent heating periods within the cycle must reheat the material and overcome the effects of decreased resistance due to deformation at the weld interface (larger surface area).

(6) The present invention recognizes and addresses the foregoing considerations, and others, of prior art constructions and methods.

SUMMARY OF THE INVENTION

(7) One embodiment of the present disclosure provides a fuel rod for a nuclear reactor including a cladding tube having a first end with an annular end face, a second end with an annular end face, and a cylindrical body portion extending therebetween, and a first tube end plug including a front portion, an annular lip with an annular end face, and a substantially straight cylindrical body portion extending therebetween, wherein the surface area of the annular end face of the first end of the cladding tube and the annular end face of the annular lip of the first tube end plug are substantially equal, and the annular end face of the first end of the cladding tube and the annular end face of the annular lip of the first tube end plug are connected by a resistance pressure weld.

(8) Another embodiment of the present disclosure provides a fuel rod for a nuclear reactor

including a cladding tube having a first end with an annular end face, a second end with an annular end face, and a cylindrical body portion extending therebetween, and a first tube end plug including a front portion, a rear face, a substantially straight cylindrical body portion extending therebetween, and an annular undercut extending from the rear face into the cylindrical body portion, the annular undercut including a bottom wall, wherein the bottom wall of the annular undercut defines a continuous curve in a plane that both includes a longitudinal center axis of, and bisects, the fuel rod, and the annular end face of the first end of the cladding tube and the rear face of the first tube end plug are connected by a resistance pressure weld.

(9) Another embodiment of the present disclosure provides a method of performing a resistance pressure weld including the steps of providing a cladding tube having a first end with an annular end face, providing a tube end plug including a front portion, a rear face, a cylindrical body portion extending therebetween, and an annular undercut extending from the rear face into the cylindrical body portion, thereby forming an annular lip having an annular end face, securing a first electrode to the first end of the cladding tube, securing a second electrode to the tube end plug, abutting the annular end face of the cladding tube to the annular end face of the tube end plug, applying compressive force to both the cladding tube and the tube end plug, and applying electric current between the first and the second electrodes so that current flows between the cladding tube and the tube end plug.

(10) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate one or more embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present invention, including the best mode thereof to one of ordinary skill in the art, is set forth more particularly in the remainder of the specification, which makes reference to the accompanying figures, in which:

(2) FIG. 1 is a diagrammatic illustration of a fuel rod including upper and lower tube end plugs in accordance with an embodiment of the present disclosure;

(3) FIG. 2 is a diagrammatic illustration of a nuclear reactor core including a plurality of fuel rods as shown in FIG. 1;

(4) FIG. 3 is a diagrammatic illustration of a nuclear reactor including the nuclear reactor core as shown in FIG. 2;

(5) FIG. 4A is a perspective view of a tube end plug in accordance with an embodiment of the present disclosure;

(6) FIG. 4B is a rear view of the tube end plug as shown in FIG. 4A;

(7) FIG. 4C is a cross-sectional view of the tube end plug as shown in FIG. 4B, taken along line 4C-4C;

(8) FIG. 4D is an enlarged cross-sectional view of a portion of the tube end plug as shown in FIG. 4C;

(9) FIG. 5 is a cross-sectional side view of a fuel rod cladding tube and tube end plug as shown in FIG. 4A, undergoing a pressure resistance welding operation;

(10) FIG. 6 is a cross-sectional view of the welded fuel rod cladding tube and tube end plug as shown in FIG. 5;

(11) FIG. 7A is a perspective view of an alternate embodiment of a tube end plug in accordance with the present disclosure;

(12) FIG. 7B is a cross-sectional view of the alternate embodiment of the tube end plug as shown in FIG. 7A;

- (13) FIGS. **8A** and **8B** are partial, cross-sectional views of the abutment region of alternate embodiments of cladding tubes and tube end plugs in accordance with the present disclosure;
- (14) FIG. **9** is a partial cross-sectional view of the welded fuel rod shown in FIG. **6**, showing the bond line between the fuel rod cladding tube and the tube end plug;
- (15) FIG. **10** is a cross-sectional view of a prior art tube end plug; and
- (16) FIG. **11** is a flow diagram of a method of producing a resistance pressure weld in accordance with the embodiment of the invention shown in FIG. **5**.
- (17) Repeat use of reference characters in the present specification and drawings is intended to represent same or analogous features or elements of the invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(18) Reference will now be made in detail to presently preferred embodiments of the invention, one or more examples of which are illustrated in the accompanying drawings. Each example is provided by way of explanation, not limitation of the invention. In fact, it will be apparent to those skilled in the art that modifications and variations can be made in the present invention without departing from the scope or spirit thereof. For instance, features illustrated or described as part of one embodiment may be used on another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

(19) Referring now to the figures, a fuel rod **12** constructed in accordance with the present disclosure is shown. Fuel rod **12** includes a cladding tube **14** containing a stack of fissile material fuel pellets **16**, and cladding tube **14** is plugged at its upper and lower ends with lower and upper tube end plugs **18** and **20**, respectively. The plugging of the lower end by lower end plug **18**, and the upper end by upper end plug **20** includes a resistance weld as disclosed herein. Fuel rod **12** may optionally contain other elements such as spacers not containing fissile material, or containing a reduced concentration of fissile material as compared with the fuel pellets.

(20) Referring now to FIG. **2**, a nuclear reactor core **10** is diagrammatically shown, and includes an array of fuel rods **12**. While the illustrative reactor core **10** includes only a 10×10 array of 100 fuel rods, a reactor for generating electrical power may employ thousands or tens of thousands of fuel rods **12**, typically arranged in structural groups called fuel assemblies. For example, one contemplated small modular reactor (SMR) design may include up to 69 fuel assemblies each comprising a 17×17 bundle of fuel rods. The fuel rods of each fuel assembly are typically held together by spacer grids welded with guide tubes and upper and lower nozzles or end plates to form the structural skeleton, and a core basket core former, or other structural support contains the fuel assemblies (structural support components not shown).

(21) Referring additionally to FIG. **3**, a nuclear reactor **11** is diagrammatically shown, and includes nuclear reactor core **10** disposed in a pressure vessel **22** and immersed in primary coolant water. In the illustrative reactor of FIG. **3**, a central riser structure **24** defines primary coolant flow circulation paths **26** which may be driven by natural circulation (that is, convection currents due to heating by reactor core **10**) or by internal or external reactor coolant pumps (not shown). Circulation paths **26** entail primary coolant heated by the reactor core **10** flowing upward through the interior plenum of the central riser **24** (the “hot” leg) and back downward to return to the core **10** via a downcomer annulus defined between the pressure vessel **22** and central riser **24** (the “cold” leg). The nuclear reactor optionally includes various other components not shown in FIG. **3**, such as control rods and associated control rod drive mechanisms (CRDMs), optional reactor coolant pumps, an internal or external pressurizer, coolant makeup and letdown sub-systems, emergency core cooling systems, an external containment structure, and so forth.

(22) Referring now to FIGS. **4A** through **4D**, a tube end plug in accordance with the present disclosure is shown. For ease of description, only bottom end plug **18** is presently discussed. As shown, tube end plug **18** including a front portion **28** with a conical outer surface, a rear face **36**, a cylindrical main body portion **30** extending therebetween, and an annular undercut **38** that depends

inwardly from rear face **36**. Preferably, annular undercut **38** is concentric with the cylindrical outer surface of main body portion **30** such that an annular lip **43** and a cylindrical fuel seat **34** are formed in main body portion **30**. As shown, fuel seat **34** is defined by annular lip **43** and a portion of rear face **36** of tube end plug **18**, and serves as a stop for supporting the stack of fuel pellets **16** (FIG. **1**) that are disposed within fuel rod **12**. Referring additionally to FIGS. **5** and **6**, annular lip **43** includes an annular end face **32** that serves as a bonding surface with a correspondingly shaped bonding surface, or annular end face **15**, of cladding tube **14**, as discussed in greater detail below. (23) Referring specifically to FIG. **4D**, annular undercut **38** preferably includes a cylindrical first inner wall **40**, a cylindrical second inner wall **42**, and a curved bottom wall **46** extending therebetween. First inner wall **40** and second inner wall **42** extend inwardly from rear face **36** of tube end plug **18**, each terminating at a corresponding inner and outer peripheral edge of bottom wall **46**. Both first inner wall **40** and second inner wall **42** are concentric to each other as well as to the cylindrical outer surface of main body portion **30**, which is also concentric about longitudinal center axis **19**. As such, first inner wall **40** and second inner wall **42** are substantially “straight,” meaning they are not angled, slanted, or oblique with respect to the longitudinal center axis beyond known manufacturing tolerances. Bottom wall **46** is a surface that forms a continuous curve between first inner wall **40** and second inner wall **42**. Moreover, as shown in FIG. **4D**, the cross-sectional shape of bottom wall **46**, in a plane that bisects tube end plug **18**, is defined by a constant radius of curvature (R). This construction minimizes the potential for areas of stress concentration in the weld region of tube end plug **18** and cladding tube **14** as no sharp corners are defined within annular undercut **38**.

(24) In order to promote uniform heating of the portions of tube end plug **18** and cladding tube **14** that are within the resistance pressure weld region, annular end face **32** of tube end plug **18** and annular end face **15** of cladding tube **14** have both substantially the same shape and surface area. Note, however, in alternate embodiments the surface area of annular end face **32** of tube end plug **18** is greater than the surface area of annular end face **15** of cladding tube **14**. This can be advantageous from a heat dissipation standpoint where one electrode is “hotter” than the other one. In those cases, the hotter electrode is frequently attached to the tube end plug because of its overall greater mass in the vicinity of the weld. The wider annular lip **43** provides a greater amount of material immediately adjacent the weld to dissipate heat on the plug side. In the embodiment shown, both annular end face **32** and annular end face **15** lie in a plane that is transverse to longitudinal center axis **19** of fuel rod **12**. Additionally, both tube end plug **18** and cladding tube **14** are comprised of martensitic stainless steel, although other materials such as, but not limited to, 304 stainless steel, ODS alloys, etc., may be used in their construction.

(25) With reference to FIGS. **5** and **6**, an illustrative resistance pressure welding process is shown for welding lower tube end plug **18** to cladding tube **14**. As noted above, lower end plug **18** preferably includes cone-shaped front end portion **28**, cylindrical main body portion **30**, as well as annular lip **43** and fuel seat **34**, both of which are defined in part by annular undercut **38**. Note, however, in alternate embodiments front end portion **28** may be rounded, flat, oblong, etc., rather than cone-shaped in cross-section. Annular lip **43** includes annular end face **32**, which has both substantially the same shape and surface area as annular end face **15** of cladding tube **14**. In short, annular end face **32** of tube end plug **18** has substantially the same inner diameter and outer diameter as does annular end face **15** of cladding tube **14**, within known manufacturing tolerances for the assembly of fuel rods. Preferably, the outer diameters of cladding tube **14** and annular lip **43** of tube end plug **18** are in the range of 0.2” to 0.6” (5.1 mm to 15.2 mm) in diameter, although welding larger tubes is possible with higher currents.

(26) Referring now to FIGS. **5** and **11**, a method of performing a resistance pressure weld between tube end plug **18** and cladding tube **14** is discussed. First, at step **80**, a cladding tube **14** and a tube end plug **18** including an annular undercut **38** in accordance with the present disclosure are provided. Next, a first electrode **50** and a second electrode **52** are attached to each of cladding tube

14 and tube end plug **18**, respectively, at step **82**. As shown in FIG. 5, first electrode **50** is a clamshell-type electrode that clamps onto the end of cladding tube **14**. In an alternate configuration, first electrode **50** can be sized to engage the end of cladding tube **14** with an interference fit between the first electrode and cladding tube **14**, thereby securing cladding tube **14** in first electrode **50**. Second electrode **52** can include a divot or other mating recess **54** that is configured to receive the front end portion **28** of lower end plug **18**, which in the instant case is conical.

(27) With cladding tube **14** and lower end plug **18** attached to their respective first and second electrodes **50** and **52**, annular end face **15** of cladding tube **14** is clamped against annular end face **32** of lower end plug **18** at step **84**. The clamping force is diagrammatically indicated in FIG. 5 by arrows (F). With the clamping force (F) applied, an electrical current is applied between first and second electrodes **50** and **52** that flows between cladding tube **14** and the lower end plug **18** while cladding tube **14** and end plug **18** are compressively loaded (step **86**). Electrical resistance is expected to be highest at the interface of annular end faces **15** and **32** of cladding tube **14** and lower end plug **18**, respectively. The resistance causes resistive heating of the mated annular end faces **15** and **32**. The combination of the clamping force (F) and the applied electric current is effective to generate a pressure resistance weld **60** (FIG. 6) between the annular end faces **15** and **32** of cladding tube **14** and lower end plug **18**. As best seen in FIG. 9, in the vicinity of pressure resistance weld **60**, inner diameter and outer diameter weld upset of both cladding tube **14** and annular lip **43** of tube end plug **18** occurs, resulting in a bond line length (along dotted line **61**) that is greater than the original interface length between end faces **32** and **15**. To achieve this effect, the clamping force (F) is approximately 200-800 pounds in some embodiments, for example. The heating due to the electrical resistance at the tube end plug/cladding tube interface causes the metal to soften, thereby forming pressure resistance weld **60**. Although alternating current can be applied with first and second electrodes **50** and **52**, direct current is preferably used as it can be regulated over time to allow controlled preheating, welding, and post heating as compared to welding with alternating current.

(28) Referring now to FIGS. 7A and 7B, an embodiment of a tube end plug **18a** in accordance with an alternate embodiment of the present disclosure is shown. Tube end plug **18a** differs only from the previously discussed embodiment shown in FIGS. 4A through 4D in that annular undercut **38a** does not include cylindrical first and second inner walls **40** and **42**. Rather, annular undercut **38** is defined solely by bottom wall **46** which, as before, is a continuously curved surface extending from its inner perimeter to its outer perimeter. Moreover, bottom wall **46** preferably has a cross-sectional shape, in a plane that bisects tube end plug **18a**, that is defined by a constant radius of curvature (R). As such, similarly to the previously discussed embodiment, annular undercut **38a** does not define any sharp corners in annular lip **43** that may contribute to stress concentrations due to the pressure resistance welding process. Like the previous embodiment, however, it can be seen that the end face **32** of the annular lip **43** and fuel seat **34** of the end plug **18** are formed and separated by an annular undercut **38** defined in the first tube end plug **18** extending from a first annular location at the annular lip end face **32** to a second annular location defining an outer circumference of the fuel seat **34**, the first annular location being radially outward of the second annular location.

(29) Referring to FIG. 8A, alternate embodiments of a lower tube end plug **18b** and cladding tube **14b** in accordance with the present disclosure are shown. Tube end plug **18b** differs only from the embodiment shown in FIGS. 7A and 7B in that annular lip **43** includes an annular end face **60** that is defined by a frustoconical surface. Similarly, cladding tube **14b** includes an annular end face **62** that is defined by a correspondingly shaped frustoconical surface **62**. Similarly to the previously discussed embodiments of tube end plugs, annular end faces **60** and **62** of tube end plug **18b** and cladding tube **14b**, respectively, have surface areas that are substantially the same. Note, however, the frustoconical annular end faces **60** and **62** offer a greater surface area, and therefore area of contact between tube end plug **18b** and cladding tube **14b**, than does a pair of annular end faces that is transverse to longitudinal center axis **19** for the same wall thickness. Referring to FIG. 8A,

alternate embodiments of a lower tube end plug **18b** and cladding tube **14b** in accordance with the present disclosure are shown. Tube end plug **18b** differs only from the embodiment shown in FIGS. 7A and 7B in that annular lip **43** includes an annular end face **60** that is defined by a frustoconical surface. Similarly, cladding tube **14b** includes an annular end face **62** that is defined by a correspondingly shaped frustoconical surface **62**. Similarly to the previously discussed embodiments of tube end plugs, annular end faces **60** and **62** of tube end plug **18b** and cladding tube **14b**, respectively, have surface areas that are substantially the same. Note, however, the frustoconical annular end faces **60** and **62** offer a greater surface area, and therefore area of contact between tube end plug **18b** and cladding tube **14b**, than does a pair of annular end faces that are transverse to longitudinal center axis **19** for the same wall thickness.

(30) Referring now to FIG. 8B, alternate embodiments of a lower tube end plug **18c** and cladding tube **14c** in accordance with the present disclosure are shown. Tube end plug **18c** differs only from the embodiment shown in FIGS. 7A and 7B in that annular lip **43** includes an annular end face **60** that is defined by a frustoconical surface. Cladding tube **14c** includes an annular end face **70** that lies in a plane that is transverse to longitudinal center axis **19** of fuel rod **12**. Note, although the surface area of annular end face **60** is greater than the surface area of annular end face **70**, the wall thicknesses are substantially similar. Note, because annular end face **60** become thinner as it comes into contact with end face **70** of cladding tube **14c**, an increased amount of material from the tube end plug's annular lip **43** is displaced as welding begins, as compared to the embodiment shown in FIGS. 7A and 7B. As such, contaminants that may be present on end faces **60** and **70** are likely to be displaced. Note, in alternate embodiments, the frustoconical surface defining annular end face **60** may be oriented so as to open radially outwardly.

(31) While one or more preferred embodiments of the present invention have been described above, it should be understood that any and all equivalent realizations of the present invention are included within the scope and spirit thereof. Thus, the depicted embodiment(s) are presented by way of example only and are not intended as limitations on the present invention. For example, the annular end faces of the tube end plug and the cladding tube may be configured differently than those shown and described herein. Specifically, the end faces may have cross-sections that are V-shaped, concave, convex, polygonal, etc., and they need not be correspondingly shaped. It should be understood that aspects of the various one or more embodiments may be interchanged both in whole or in part. Therefore, it is contemplated that any and all such embodiments are included in the present invention as may fall within the literal or equivalent scope of the appended claims.

Claims

1. A fuel rod for a nuclear reactor, comprising: a cladding tube having a first end with an annular end face, a second end with an annular end face, and a cylindrical body portion extending therebetween; and a first tube end plug including a front portion, an annular lip with an annular end face, and a substantially straight cylindrical body portion extending therebetween, the annular lip and a fuel seat of the first tube end plug being formed and separated by an annular undercut defined in the first tube end plug, the annular undercut having a bottom wall that is axially positioned between the annular lip and the front portion of the first tube end plug; wherein the annular end face of the first end of the cladding tube defines a first frustoconical surface and the annular end face of the annular lip of the first tube end plug defines a second frustoconical surface, the first frustoconical surface and the second frustoconical surface being complementary so as to oppose and abut one another, and the annular end face of the first end of the cladding tube and the annular end face of the annular lip of the first tube end plug are connected by a resistance pressure weld.
2. The fuel rod of claim 1, wherein a surface area of the annular end face of the first end of the cladding tube and the annular end face of the annular lip of the first tube end plug are substantially equal.

3. The fuel rod of claim 1, wherein the annular end face of the first end of the cladding tube lies before welding in a plane that is transverse to a longitudinal axis of the cladding tube and the annular end face of the annular lip of the first tube end plug is defined as a frustoconical surface.
 4. The fuel rod of claim 1, further comprising a second tube end plug located at the second end of the cladding tube.
 5. The fuel rod of claim 4, wherein the second tube end plug is substantially identical to the first tube end plug.
 6. A fuel rod for a nuclear reactor, comprising: a cladding tube having a first end with an annular end face, a second end with an annular end face, and a cylindrical body portion extending therebetween; and a first tube end plug including a front portion, an annular lip with an annular end face, and a substantially straight cylindrical body portion extending therebetween, the annular lip and a fuel seat of the first tube end plug being formed and separated by an annular undercut defined in the first tube end plug, the annular undercut having a bottom wall that is axially positioned between the annular lip and the front portion of the first tube end plug; and the annular end face of the first end of the cladding tube and the annular end face of the annular lip of the first tube end plug are connected by a resistance pressure weld, wherein the bottom wall of the annular undercut defines a constant radius of curvature in cross section.
 7. A fuel rod for a nuclear reactor, comprising: a cladding tube having a first end with an annular end face, a second end with an annular end face, and a cylindrical body portion extending therebetween; and a first tube end plug including a front portion, an annular lip with an annular end face, and a substantially straight cylindrical body portion extending therebetween, the annular lip and a fuel seat of the first tube end plug being formed and separated by an annular undercut defined in the first tube end plug, the annular undercut having a bottom wall that is axially positioned between the annular lip and the front portion of the first tube end plug; and the annular end face of the first end of the cladding tube and the annular end face of the annular lip of the first tube end plug are connected by a resistance pressure weld, wherein the annular undercut has a cylindrical first inner wall and a cylindrical second inner wall that are connected by the bottom wall, wherein the bottom wall of the annular undercut defines a continuous curve between the cylindrical first inner wall and the cylindrical second inner wall.
 8. A fuel rod for a nuclear reactor, comprising: a cladding tube having a first end with an annular end face, a second end with an annular end face, and a cylindrical body portion extending therebetween; and a first tube end plug including a front portion, an annular lip with an annular end face, and a substantially straight cylindrical body portion extending therebetween, the annular lip end face and a fuel seat of the first tube end plug being formed and separated by an annular undercut defined in the first tube end plug extending from a first annular location at the annular end face of the annular lip to a second annular location defining an outer circumference of the fuel seat, the first annular location being radially outward of the second annular location, the annular undercut having a bottom wall that is axially positioned between the annular lip and the front portion of the first tube end plug; and the annular end face of the first end of the cladding tube and the annular end face of the annular lip of the first tube end plug are connected by a weld.
 9. The fuel rod of claim 8, wherein the annular end face of the annular lip and a surface of the fuel seat are in a common plane transverse to an axis of the first tube end plug.
 10. The fuel rod of claim 8, wherein at least one of the annular end face of the annular lip of the first tube end plug and the annular end face of the first end of the cladding tube defines a frustoconical surface.
 11. The fuel rod of claim 10, wherein the annular end face of the first end of the cladding tube defines a first frustoconical surface and the annular end face of the annular lip of the first tube end plug defines a second frustoconical surface, the first frustoconical surface and the second frustoconical surface being complementary so as to oppose and abut one another.
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